

Logistic Regression (2)

(로지스틱 회귀분석 (2))

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Today's Topics

1. Logistic Regression

- Influential Diagnostics
- Logistic Regression for Qualitative Variables
- $2 \times 2 \times K$ Contingency Table
- Multiple Logistic Regression
- Model Selection
- Predictive Power
- Effects of Sparse Data
- Adjusting sampling bias in oversampled logistic modeling
- Exact test for logistic regression

Influence Diagnostics

Measure

The presence of influential point affect the regression result

Diagnostic of influential point

1. DFBETA: observations with larger than $2\sqrt{(p+1)/n}$, can be influential points

$$\frac{\hat{\beta} - \hat{\beta}_{(-i)}}{\sqrt{\text{var}(\hat{\beta} - \hat{\beta}_{(-i)})}}$$

$\hat{\beta}_{(-i)}$: regression coefficient estimate when the observation i is removed

2. Cook's Distance: observations with larger than 1 can be influential points

$$\frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(-i)})^2}{(p+1)\hat{\sigma}^2}$$

3. Leverage: amount of deviation in terms of predictor. If it is larger than $(p+1)/n$, then "high leverage"

Influence Diagnostics (Cont'd)

(Practice) Horseshoe crabs data

```
> results_table
```

	DFBETA	Cooks_D	Leverage
23	0.6710666	0.04334754	0.02571721
128	0.6710666	0.04334754	0.02571721
54	0.5869870	0.03192985	0.02322997
72	0.5651846	0.02944252	0.02254400
21	0.5209212	0.02489198	0.02111417

```
## Horseshoe crab data
```

```
require(glm2)
```

```
data(crabs)
```

```
crabs$HasSatellite <- ifelse(crabs$Satellites > 0, 1, 0)
```

```
model_logistic <- glm(HasSatellite ~ Width, family =  
binomial(link = "logit"), data = crabs)
```

```
summary(model_logistic)
```

```
# Influence of Diagnostics
```

```
dfbeta_values <- dfbeta(model_logistic)
```

```
cooks_d <- cooks.distance(model_logistic)
```

```
leverage_values <- hatvalues(model_logistic)
```

```
top_indices <- head(order(dfbeta_values, decreasing =  
TRUE), 5)
```

```
results_table <- data.frame(DFBETA =  
dfbeta_values[top_indices], Cooks_D = cooks_d[top_indices],  
Leverage = leverage_values[top_indices]); results_table
```

Logistic Regression for Qualitative Variables

Dummy variables

Main effect model

Binary explanatory variables: X, Z , binary response variable: Y

Regression model without interaction

$$\text{logit}(\pi) = \beta_0 + \beta_1 x + \beta_2 z$$

The effect of one variable are not affected by the level of the other variables

Difference of logit between $x = 0$ and 1

$$[\alpha + \beta_1(1) + \beta_2 z] - [\alpha + \beta_1(0) + \beta_2 z] = \beta_1$$

Conditional odds ratio = $\exp(\beta_1)$: same for different values of z

That is, common odds ratio

Logistic Regression for Qualitative Variables (Cont'd)

(Practice) AZT and AIDS data

Data

Race (Z)	AZT use (X)	Worsening of symptoms	
		Yes	No
White (1)	Yes (1)	14	93
	No (0)	32	81
Black (0)	Yes (1)	11	52
	No (0)	12	43

Output

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.07357	0.26294	-4.083	4.45e-05	***
azt	-0.71946	0.27898	-2.579	0.00991	**
race	0.05548	0.28861	0.192	0.84755	

```
## AZT and AIDS data
```

```
aids <- data.frame(  
  race = c(1, 1, 0, 0),  
  azt = c(1, 0, 1, 0),  
  yes = c(14, 32, 11, 12),  
  no = c(93, 81, 52, 43)  
)
```

```
aids_model_logistic <- glm(cbind(yes, no) ~ azt+race,  
  data=aids, family=binomial(link="logit"))  
summary(aids_model_logistic)
```

Logistic Regression for Qualitative Variables (Cont'd)

(Practice) AZT and AIDS data

Interpretation

$$\hat{\beta}_1 = -0.7195(0.2790), P = 0.0099$$

Odds ratio between AZT and AIDS is $e^{-0.7195} = 0.49$

That is, for each race, the conditional odds at which symptoms would worsen were about $\frac{1}{2}$ lower in those who received AZT immediately than those who did not

$$\hat{\beta}_2 = 0.0555(0.2886), P = 0.8476$$

The effect of race is not significant. It is better to fit the model without race variable

$2 \times 2 \times K$ Contingency Table

Logistic regression

Z : control variable with K levels

If $\beta_1^X = 0$, X and Y are conditionally independent after adjusting Z

If X and Y are conditionally independent, the effect of X and β_i^X , can be included in β_0 :

$$\text{logit}(\pi) = \beta_0 + \beta_1^Z z_1 + \dots + \beta_{K-1}^Z z_{K-1}$$

That is, the model without X can be fitted

$2 \times 2 \times K$ Contingency Table (Cont'd)

Conditional independence

Likelihood ratio: $-2[\log(L_1) - \log(L_2)]$

CMH test

It performs well if the odds ratios for each partial contingency table are similar

Common odds ratio

Logistic model

Test: likelihood ratio test $-2[\log(L_1) - \log(L_2)]$

Parameter estimation: maximum likelihood estimator $\exp(\hat{\beta}_1^X - \hat{\beta}_0^X)$

Contingency table

Test: Breslow-Day test

Parameter estimation: Mantel-Haenszel common odds ratio

$2 \times 2 \times K$ Contingency Table (Cont'd)

(Practice) AZT and AIDS data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.07357	0.26294	-4.083	4.45e-05 ***
azt	-0.71946	0.27898	-2.579	0.00991 **
race	0.05548	0.28861	0.192	0.84755

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8.3499 on 3 degrees of freedom
Residual deviance: 1.3835 on 1 degrees of freedom
AIC: 24.86

Likelihood ratio test

Model 1: cbind(yes, no) ~ race

Model 2: cbind(yes, no) ~ azt + race

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	2	-12.8654			
2	3	-9.4299	1	6.8709	0.008761 **

```
aids_model_logistic <- glm(cbind(yes, no) ~ azt+race,  
data=aids, family=binomial(link="logit"))  
summary(aids_model_logistic)
```

```
aids_model_logistic_red <- glm(cbind(yes, no) ~ race,  
data=aids, family=binomial(link="logit"))  
summary(aids_model_logistic_red)  
lmtest::lrtest(aids_model_logistic_red, aids_model_logistic)
```

```
aids_3d <- array(data = c(14, 32, 93, 81, 11, 12, 52, 43), dim =  
c(2, 2, 2), dimnames = list( azt = c("Yes", "No"), worsening =  
c("Yes", "No"), race = c("White", "Black")))
```

```
# CMH test (Conditional Independence)  
mantelhaen.test(aids_3d)
```

```
# Breslow-Day Test (Homogeneity)  
bd <- DescTools::BreslowDayTest(aids_3d); bd
```

$2 \times 2 \times K$ Contingency Table (Cont'd)

(Practice) AZT and AIDS data

```
> mantelhaen.test(aids_3d)
```

Mantel-Haenszel chi-squared test with continuity correction

```
data: aids_3d
Mantel-Haenszel X-squared = 6.0799, df = 1, p-value = 0.01367
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 0.2818621 0.8414208
sample estimates:
common odds ratio
 0.4869955
```

```
> bd <- DescTools::BreslowDayTest(aids_3d); bd
```

Breslow-Day test on Homogeneity of Odds Ratios

```
data: aids_3d
X-squared = 1.391, df = 1, p-value = 0.2382
```

```
aids_model_logistic <- glm(cbind(yes, no) ~ azt+race,
data=aids, family=binomial(link="logit"))
summary(aids_model_logistic)
```

```
aids_model_logistic_red <- glm(cbind(yes, no) ~ race,
data=aids, family=binomial(link="logit"))
summary(aids_model_logistic_red)
lmtest::lrtest(aids_model_logistic_red, aids_model_logistic)
```

```
aids_3d <- array(data = c(14, 32, 93, 81, 11, 12, 52, 43), dim =
c(2, 2, 2), dimnames = list( azt = c("Yes", "No"), worsening =
c("Yes", "No"), race = c("White", "Black")))
```

```
# CMH test (Conditional Independence)
mantelhaen.test(aids_3d)
```

```
# Breslow-Day Test (Homogeneity)
bd <- DescTools::BreslowDayTest(aids_3d); bd
```

$2 \times 2 \times K$ Contingency Table (Cont'd)

Output

Regression coefficient estimates

$$\hat{\beta}_0 = -1.0736, \hat{\beta}_{No}^X = 0, \hat{\beta}_{Yes}^X = -0.7195, \hat{\beta}_{Black}^Z = 0, \hat{\beta}_{White}^Z = 0.0555$$

Conditional independence

Likelihood ratio test = 6.87, CMH test = 6.08

$P = 0.01$, there is sufficient evidence to reject the null hypothesis, conditional independence

Common odds ratio

Homogeneous association test: (Goodness-of-fit test) $1.3835 \sim \chi^2(df = 1)$

Conditional odds ratio MLE: $\exp(\hat{\beta}_{Yes}^X - \hat{\beta}_{No}^X) = 0.49$

Breslow-Day test: $1.3910 \sim \chi^2(df = 1)$

Mental Haenszel odds ratio: 0.49

Multiple Logistic Regression

Meaning of regression coefficient

Logistic regression with multiple explanatory variables

$$\text{logit}(\pi) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

The meaning of β_i

$\exp(\beta_i)$: odds ratio for on unit increase of X_i when the other variables are controlled

Model comparison

Likelihood ratio test

It can be used to test the hypothesis about regression coefficients for multiple logistic regression

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.7151	2.7617	-4.604	4.14e-06	***
ColDM	1.1061	0.5921	1.868	0.0617	.
ColM	1.4023	0.5484	2.557	0.0106	*
ColLM	1.3299	0.8525	1.560	0.1188	
Width	0.4680	0.1055	4.434	9.26e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 225.76 on 172 degrees of freedom
Residual deviance: 187.46 on 168 degrees of freedom
AIC: 197.46

Hosmer and Lemeshow goodness of fit (GOF) test

data: actual_response, predicted_probs
X-squared = 4.4581, df = 8, p-value = 0.8136

```
## Horseshoe crab data
```

```
require(GLMsData)
```

```
hcrabs$HasSatellite <- ifelse(hcrabs$Sat > 0, 1, 0)
```

```
hcrabs$Col <-
```

```
factor(hcrabs$Col, levels=c("D", "DM", "M", "LM"))
```

```
hcrabs_model_logistic <- glm(HasSatellite ~ Col + Width,  
family = binomial(link = "logit"), data = hcrabs)
```

```
summary(hcrabs_model_logistic)
```

```
# Hosmer-Lemeshow test
```

```
require(ResourceSelection)
```

```
predicted_probs <- fitted(hcrabs_model_logistic)
```

```
actual_response <- hcrabs$HasSatellite
```

```
hoslem.test(actual_response, predicted_probs, g = 10)
```

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

Explanatory variables: width, color

Color is a qualitative variable with four levels (D,DM,M,LM)

$$\text{logit}(\pi) = \beta_0 + \beta_1 \text{color}(DM) + \beta_2 \text{color}(M) + \beta_3 \text{color}(LM) + \beta_4 \text{width}$$

Parameter estimates:

$$\hat{\beta}_0 = -12.7151, \hat{\beta}_1 = 1.106 (p = 0.062), \hat{\beta}_2 = 1.402 (0.011), \\ \hat{\beta}_3 = 1.330(0.119), \hat{\beta}_4 = 0.468(9.26 \times 10^{-6})$$

Goodness-of-fit: $G^2 = 187.46$ ($df = 168$),

$$\hat{C} = 4.458 \sim \chi^2(df = 8), p = 0.8136$$

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

Estimated probabilities

If width=26.3cm and color=LM,
$$\frac{e^{-12.715+1.33+0.468 \times 26.3}}{1 + e^{-12.715+1.33+0.468 \times 26.3}} = 0.708$$

If width=26.3cm and color=D,
$$\frac{e^{-12.715+0.468 \times 26.3}}{1 + e^{-12.715+0.468 \times 26.3}} = 0.391$$

Odds ratios

Odds ratio between color=LM and color=D of satellite is

$$e^{1.330} = 3.78 = \frac{0.708/0.292}{0.391/0.609}$$

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-5.85383	6.69393	-0.874	0.3818
ColDM	-15.66423	9.56065	-1.638	0.1013
ColM	-4.18613	7.58093	-0.552	0.5808
ColLM	4.10122	13.27532	0.309	0.7574
Width	0.20043	0.26166	0.766	0.4437
ColDM:Width	0.65794	0.37533	1.753	0.0796
ColM:Width	0.21844	0.29524	0.740	0.4594
ColLM:Width	-0.09443	0.50042	-0.189	0.8503

```
hcrabs_model_logistic_interaction <-  
glm(HasSatellite ~ Col*Width, family =  
binomial(link = "logit"), data = hcrabs)  
summary(hcrabs_model_logistic_interaction)
```

```
lmtest::lrtest(hcrabs_model_logistic,  
hcrabs_model_logistic_interaction)
```

Likelihood ratio test

Model 1: HasSatellite ~ Col + Width

Model 2: HasSatellite ~ Col * Width

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	5	-93.729			
2	8	-91.540	3	4.3764	0.2236

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

Likelihood ratio test between models with and without interaction

$$G^2 = -2(-93.729 + 91.540) = 4.3764 \sim \chi^2(df = 3), p = 0.2236$$

Multiple Logistic Regression (Cont'd)

Qualitative variable selection

Model selection

For the analyses of qualitative explanatory variables, the best model can be selected by considering the coefficients of dummy variables

1. Consider it as a continuous variable and repeat analyses
2. Recategorize the levels and repeat analyses

If ordinal variables are considered as an continuous variable, interpretation is much easier

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Idea 1

Model 2: Levels of color can be ordered by the darkness of each color

→ can be considered as a continuous variable ($c = \{1,2,3,4\}$)

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.6161	2.6868	-4.696	2.66e-06	***
Col_conti	0.5090	0.2237	2.276	0.0229	*
Width	0.4583	0.1040	4.406	1.05e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 225.76 on 172 degrees of freedom
Residual deviance: 189.12 on 170 degrees of freedom
AIC: 195.12

Hosmer and Lemeshow goodness of fit (GOF) test

data: actual_response, predicted_probs
X-squared = 9.7791, df = 8, p-value = 0.2809

Model 1: HasSatellite ~ Col_conti + Width

Model 2: HasSatellite ~ Col + Width

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	3	-94.561			
2	5	-93.729	2	1.6641	0.4351

```
# model selection of qualitative variable
hcrabs$Col_conti <-
ifelse(hcrabs$Col=="D",1,ifelse(hcrabs$Col=="DM",2,ifelse(
hcrabs$Col=="M",3,4)))
hcrabs_model_logistic_conti <- glm(HasSatellite ~
Col_conti + Width, family = binomial(link = "logit"), data
= hcrabs)
summary(hcrabs_model_logistic_conti)

predicted_probs <- fitted(hcrabs_model_logistic_conti)
actual_response <- hcrabs$HasSatellite
hoslem.test(actual_response, predicted_probs, g = 10)

lmtest::lrtest(hcrabs_model_logistic_conti,
hcrabs_model_logistic)
```

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

Parameter estimates: $\hat{\beta}_0 = -12.616$, $\hat{\beta}_1 = 0.509$ ($p = 0.023$), $\hat{\beta}_2 = 0.458$ (1.05×10^{-5})

If the color becomes one unit darker, odds ratio becomes $e^{-0.509} = 0.6$

Goodness-of-fit: $G^2 = 187.12$ ($df = 170$),

$$\hat{C} = 9.779 \sim \chi^2(df = 8), p = 0.2809$$

Compare it with the model using dummy variables

$$G^2 = -2(-94.561 + 93.729) = 1.664 \sim \chi^2(df = 2), p = 0.4351$$

→ Under the assumption that the model using dummy variables is appropriate, it can determine whether the model using it as a continuous variable is sufficiently good

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Idea 2

Model 3: When color = DM, M, LM, and their effects are similar

→ If $c = \{DM, M, LM\}$, color=1, and otherwise color=0

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-12.9795	2.7272	-4.759	1.94e-06	***
Col_bin	1.3005	0.5259	2.473	0.0134	*
Width	0.4782	0.1041	4.592	4.39e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 225.76 on 172 degrees of freedom
Residual deviance: 187.96 on 170 degrees of freedom
AIC: 193.96

Hosmer and Lemeshow goodness of fit (GOF) test

data: actual_response, predicted_probs
X-squared = 4.0228, df = 8, p-value = 0.8551

Model 1: HasSatellite ~ Col_bin + Width

Model 2: HasSatellite ~ Col + Width

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	3	-93.979			
2	5	-93.729	2	0.5008	0.7785

```
hcrabs$Col_bin <- ifelse(hcrabs$Col=="D",0,1)
hcrabs_model_logistic_bin <- glm(HasSatellite ~
Col_bin + Width, family = binomial(link = "logit"),
data = hcrabs)
summary(hcrabs_model_logistic_bin)

predicted_probs <- fitted(hcrabs_model_logistic_bin)
actual_response <- hcrabs$HasSatellite
hoslem.test(actual_response, predicted_probs, g = 10)

lmtest::lrtest(hcrabs_model_logistic_bin,
hcrabs_model_logistic)
```


Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

For Model 1, the parameter estimates for dummy variables are $\{0, 1.11, 1.40, 1.33\}$, and the first three are similar. Thus we can consider the different coding as follows:

→ Recode color as $\{0, 1, 1, 1\}$

$$\hat{\beta}_0 = -12.980, \hat{\beta}_1 = 1.301 (p = 0.013), \hat{\beta}_2 = 0.478 (4.39 \times 10^{-6})$$

Compare it with the model using dummy variables

$$G^2 = -2(-93.979 + 93.729) = 0.5008 \sim \chi^2(df = 2), p = 0.7785$$

$$\text{AIC}(\text{Model 1}) = 197.46, \text{AIC}(\text{Model 2}) = 195.12, \text{AIC}(\text{Model 3}) = 193.96$$

→ Model 3 is better than Model 1 and 2

Multiple Logistic Regression (Cont'd)

Other comments

Sample size

(Rough guideline) The number of subjects should be larger than (the number of variables) $\times 10$ for each level of Y

For example, if there are 1000 subjects who consist of 30 subjects with $Y = 1$ and 970 subjects with $Y = 0$, the reasonable number of explanatory variable is 3

Multicollinearity

Under the multicollinearity, that is, there is high correlation between explanatory variables, $(X^t X)$ can be singular and significant variables can be non-significant

ASE becomes larger and coefficients can have weird value

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-9.6736792	3.8645627	-2.503	0.01231	*
Width	0.2631279	0.1952986	1.347	0.17788	
Wt	0.0008258	0.0007038	1.173	0.24069	
ColDM	1.1198016	0.5932901	1.887	0.05910	.
ColM	1.5057635	0.5666668	2.657	0.00788	**
ColLM	1.6086658	0.9355326	1.720	0.08552	.
SpineNoneOK	0.4002868	0.5027043	0.796	0.42588	
SpineOneOK	-0.0959809	0.7033698	-0.136	0.89146	

Model 1: HasSatellite ~ 1

Model 2: HasSatellite ~ Width + Wt + Col + Spine

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	1	-112.879			
2	8	-92.601	7	40.556	9.848e-07 ***

```
> car::vif(hcrabs_model_logistic_multi)
```

	GVIF	Df	GVIF^(1/(2*Df))
Width	3.494300	1	1.869305
Wt	3.395710	1	1.842745
Col	1.294381	3	1.043944
Spine	1.339812	2	1.075873

```
# Multicollinearity
```

```
hcrabs_model_logistic_only_intercept <-  
glm(HasSatellite ~ 1, family = binomial(link = "logit"),  
data = hcrabs)  
summary(hcrabs_model_logistic_only_intercept)
```

```
hcrabs_model_logistic_multi <- glm(HasSatellite ~  
Width+Wt+Col+Spine, family = binomial(link = "logit"),  
data = hcrabs)  
summary(hcrabs_model_logistic)
```

```
lmtest::lrtest(hcrabs_model_logistic_only_intercept,hcrabs  
_model_logistic)
```

```
cor(hcrabs$Width,hcrabs$Wt)
```

```
car::vif(hcrabs_model_logistic_multi)
```

Multiple Logistic Regression (Cont'd)

(Practice) Horseshoe crabs data

Interpretation

Likelihood ratio test: $G^2 = -2(-112.879 + 92.601) = 40.556 \sim \chi^2(df = 7), p = 9.848 \times 10^{-7}$

→ One or more variables have significant coefficients

Wald test for Model 2

P-values of most variables are larger than 0.05, which is inconsistent with LRT

→ This result indicates evidence of multicollinearity

Correlation between width and weight = 0.887: multicollinearity must be adjusted

→ Only width can be considered as a covariate or PCA can be applied to reduce the number of variables

Model Selection

Model selection method

Nested models

For M_0 and M_1 where $M_0 \subset M_1$, M_0 is called nested in M_1

Model selection with backward, forward, or stepwise based on likelihood ratio test can be used

Nonnested models

AIC, BIC, etc

Example

Check the relationship among three models

Model 1: $g(\pi_i) = \beta_0 + \beta_1 x_{1i}$

Model 2: $g(\pi_i) = \beta_0 + \beta_1 x_{2i}$

Model 3: $g(\pi_i) = \beta_0 + \beta_1 x_{1i} + \beta_1 x_{2i}$

Model Selection (Cont'd)

Akaike information creiterion (AIC)

Goal of variable selection

Identify the model which explains the response variable with relatively small number of variables (parsimonious model), not correct model

AIC

Instead of likelihood ratio test based on P-value, it determines the best model with the smallest value. It is defined by the sum of log-likelihood and penalty as follows:

$AIC = -2(\log L - \text{Penalty})$; $AIC = -2\log L + 2k$; $BIC = -2\log L + k\log n$; $BICC = -2\log L + k(\log n + 1)$

Model selection

Choose the model with the smallest AIC

Model Selection (Cont'd)

Backward elimination procedure

Property of backward elimination

It start from the most complicated model and remove one or more variables from the previous model at each step

In each step, the variable with the lowest AIC for $H_0: \beta = 0$ is selected

For each variable, the polynomial with the largest order is tested

For the model with several interaction terms, then interaction term with the highest order is only tested

Model Selection (Cont'd)

(Practice) Horseshoe crabs data

Start: AIC=212.44

HasSatellite ~ Col * Width * Spine

	Df	Deviance	AIC
- Col:Width:Spine	3	173.67	209.67
<none>		170.44	212.44

Step: AIC=209.67

HasSatellite ~ Col + Width + Spine + Col:Width + Col:Spine + Width:Spine

	Df	Deviance	AIC
- Col:Spine	6	181.56	205.56
- Width:Spine	2	173.68	205.68
- Col:Width	3	177.34	207.34
<none>		173.67	209.67

Step: AIC=205.56

HasSatellite ~ Col + Width + Spine + Col:Width + Width:Spine

	Df	Deviance	AIC
- Width:Spine	2	181.64	201.64
- Col:Width	3	186.41	204.41
<none>		181.56	205.56

```
# Backward Elimination
```

```
hcrabs_model_logistic_full <- glm(HasSatellite ~  
Col*Width*Spine, family = binomial(link = "logit"),  
data = hcrabs)
```

```
hcrabs_model_logistic_null <- glm(HasSatellite ~ 1,  
family = binomial(link = "logit"), data = hcrabs)
```

```
backward_model <- step(hcrabs_model_logistic_full,  
direction = "backward", scope =  
list(lower=formula(hcrabs_model_logistic_null),upper=f  
ormula(hcrabs_model_logistic_full)), trace = 1)
```

```
forward_model <- step(hcrabs_model_logistic_null,  
direction = "forward", scope  
=list(lower=formula(hcrabs_model_logistic_null),upper  
=formula(hcrabs_model_logistic_full)), trace = 1)
```

```
stepwise_model <- step(hcrabs_model_logistic_full,  
direction = "both", scope  
=list(lower=formula(hcrabs_model_logistic_null),upper  
=formula(hcrabs_model_logistic_full)), trace = 1)
```


Model Selection (Cont'd)

(Practice) Horseshoe crabs data

Step: AIC=201.64

HasSatellite ~ Col + Width + Spine + Col:Width

	Df	Deviance	AIC
- Spine	2	183.08	199.08
- Col:Width	3	186.61	200.61
<none>		181.64	201.64

Step: AIC=199.08

HasSatellite ~ Col + Width + Col:Width

	Df	Deviance	AIC
- Col:Width	3	187.46	197.46
<none>		183.08	199.08

Step: AIC=197.46

HasSatellite ~ Col + Width

	Df	Deviance	AIC
<none>		187.46	197.46
- Col	3	194.45	198.45
- Width	1	212.06	220.06

```
# Backward Elimination
```

```
hcrabs_model_logistic_full <- glm(HasSatellite ~  
Col*Width*Spine, family = binomial(link = "logit"),  
data = hcrabs)
```

```
hcrabs_model_logistic_null <- glm(HasSatellite ~ 1,  
family = binomial(link = "logit"), data = hcrabs)
```

```
backward_model <- step(hcrabs_model_logistic_full,  
direction = "backward", scope =  
list(lower=formula(hcrabs_model_logistic_null),upper=f  
ormula(hcrabs_model_logistic_full)), trace = 1)
```

```
forward_model <- step(hcrabs_model_logistic_null,  
direction = "forward", scope  
=list(lower=formula(hcrabs_model_logistic_null),upper  
=formula(hcrabs_model_logistic_full)), trace = 1)
```

```
stepwise_model <- step(hcrabs_model_logistic_full,  
direction = "both", scope  
=list(lower=formula(hcrabs_model_logistic_null),upper  
=formula(hcrabs_model_logistic_full)), trace = 1)
```

Predictive Power

Predicted value

How to determine $\hat{y}_i$:

1. Calculate the probability of subject i , $\hat{\pi}_i$
2. For some constant, π_0 , $\hat{y}_i$ can be determined as follows:

If $\hat{\pi}_i > \pi_0$, $\hat{y}_i = 1$, and If $\hat{\pi}_i \leq \pi_0$, $\hat{y}_i = 0$

Result can be differ according to π_0

Summary measure of predictive power

Measures which can be used to evaluate predictor power:

Sensitivity: $P(\hat{y} = 1|y = 1)$

Specificity: $P(\hat{y} = 0|y = 0)$

Correct classifications: $P((\hat{y} = 0, y = 0) \text{ or } (\hat{y} = 1, y = 1))$

ROC curve / AUC

Correlation between y_i and $\hat{\pi}_i$

Predictive Power (Cont'd)

Example

For $\pi_0 = 0.64$ and $\pi_0 = 0.5$, calculate the sensitivity, specificity, and correct classification rates

Actual	Predictino, $\pi_0 = 0.64$		Predictino, $\pi_0 = 0.50$		Total
	$\hat{y}=1$	$\hat{y}=0$	$\hat{y}=1$	$\hat{y}=0$	
y=1	74	37	94	17	111
y=0	20	42	37	25	62

For $\pi_0 = 0.64$,

Sensitivity: $74/111=0.667$, Specificity: $42/62=0.667$

Correct classification rates: $(74+42)/173=0.671$

For $\pi_0 = 0.50$,

Sensitivity: $94/111=0.847$, Specificity: $25/62=0.390$

Correct classification rates: $(94+25)/173=0.688$

Seneitivity, and specificity are $(1,0)$ for $\pi_0 = 0$, and $(0,1)$ for $\pi_0 = 1$

Predictive Power (Cont'd)

Receiver operating characteristic (ROC) curve

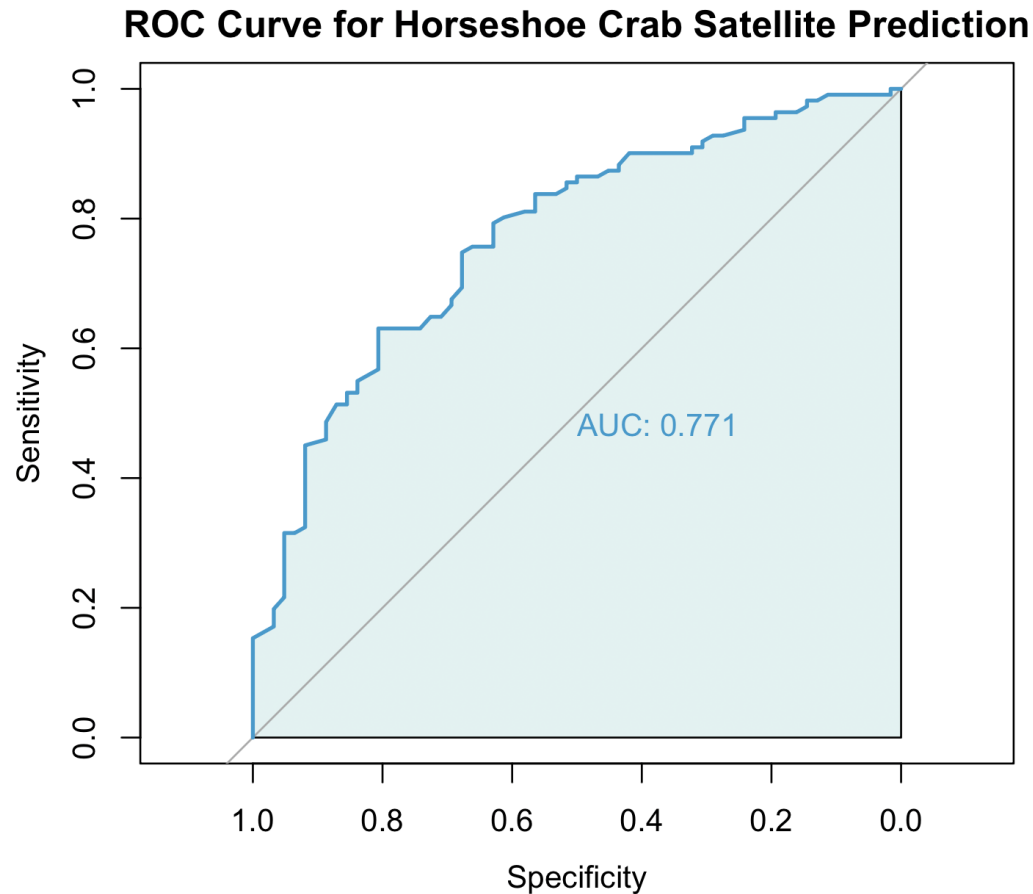
It is a graph which connects (1-specificity, sensitivity) calculated with different value of π_0

Area under curve (AUC): area of ROC curve

AUC indicates the probability of concordance index, that is, the probability that the predicted results and true results are the same

Predictive Power (Cont'd)

(Practice) Horseshoe crabs data



```
# ROC curve  
require(pROC)
```

```
hcrabs_model_logistic <- glm(HasSatellite ~ Col + Width,  
family = binomial(link = "logit"), data = hcrabs)  
probabilities <- predict(hcrabs_model_logistic, type =  
"response")
```

```
roc_obj <- roc(hcrabs$HasSatellite, probabilities)  
auc_value <- auc(roc_obj)  
plot(roc_obj, main = "ROC Curve for Horseshoe Crab  
Satellite Prediction", col = "#1C9BCF", lwd = 2, print.auc =  
TRUE, auc.polygon = TRUE, auc.polygon.col = "#E0F2F1")
```

```
hcrabs_model_logistic_null <- glm(HasSatellite ~ 1, family =  
binomial(link = "logit"), data = hcrabs)  
probabilities_null <- predict(hcrabs_model_logistic_null, type =  
"response")  
roc_obj_null <- roc(hcrabs$HasSatellite, probabilities_null)
```

```
delong_result <- roc.test(roc_obj, roc_obj_null, method =  
"delong")
```

Predictive Power (Cont'd)

(Practice) Horseshoe crabs data

```
> delong_result
```

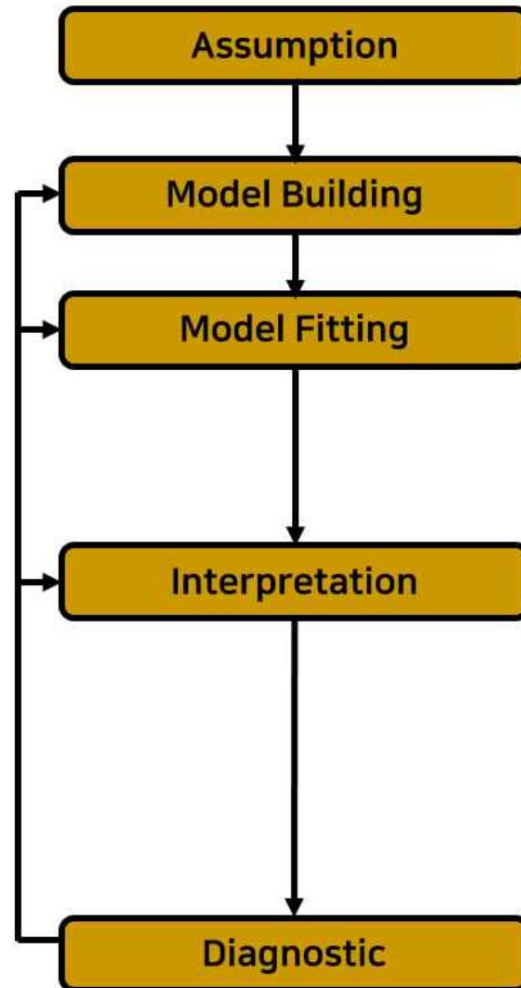
DeLong's test for two correlated ROC curves

```
data:  roc_obj and roc_obj_null
Z = 7.4129, p-value = 1.236e-13
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 0.1996123 0.3431078
sample estimates:
AUC of roc1 AUC of roc2
 0.7713601  0.5000000
```

Interpretation

AUC is 0.7714 and significantly larger than 0.5

Analysis Procedure



GLM Model

- Distribution: binomial, Poisson, Normal, etc
- Link function: logit, log, etc

Setting predictor/response variables

- Contingency table, scatter plot
- Variable transformation

MLE

- Newton Raphson, Scoring method

Goodness of Fit

- Group data: deviance / pearson-chisquare
- Ungroup data: Hosmer-Lemeshow statistics

Regression coefficients

- Wald/likelihood-ratio test

Model selection

- Nested model: likelihood ratio test
- Nonnested model: AIC, BIC, etc

Multicollinearity

Outliers/influential points

- Residuals
- Cooks Dist
- DFBETA

Prediction accuracy

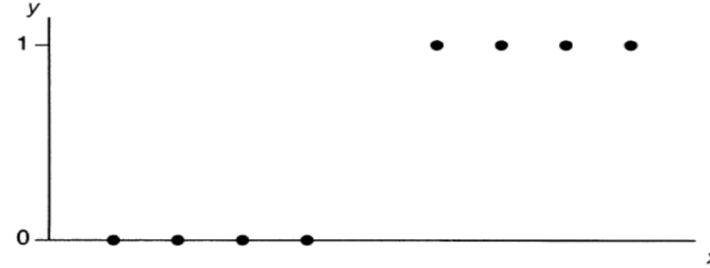
- AUC, Balanced accuracy

Effects of Sparse Data

Infinite Effect Estimate

Quantitative Predictor

If subjects with $y = 0$ and $y = 1$ can be completely classified by the some covariates, then the estimate of β becomes ∞ or $-\infty$



Categorical Predictor

If there exist some empty cells, the corresponding MLE of β is either ∞ or $-\infty$

For example, consider two explanatory variables X_1, X_2 , and Y . Then if the cell counts for marginal contingency table of X_1, Y are 0, the corresponding coefficient estimates of X_1 are ∞ or $-\infty$

Solution

If we add the small value (10^{-8}) to each cell, or pool the level of variables with 0 cell counts, then coefficients can be estimated. However some caution is necessary if their standard error are large

Effects of Sparse Data (Cont'd)

(Practice) Clinical Trial Data

```
> trials
```

	Center	Treatment	Success	Failure
1	1	Active drug	0	5
2	1	Placebo	0	9
3	2	Active drug	1	12
4	2	Placebo	0	10
5	3	Active drug	0	7
6	3	Placebo	0	5
7	4	Active drug	6	3
8	4	Placebo	2	6
9	5	Active drug	5	9
10	5	Placebo	2	12

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
Center1	-23.0462	23296.3959	-0.001	0.9992
Center2	-2.6565	1.0360	-2.564	0.0103 *
Center3	-23.0414	21523.6453	-0.001	0.9991
Center4	0.5868	0.6053	0.969	0.3323
Center5	-0.4763	0.5058	-0.942	0.3463
TreatmentPlacebo	-1.5460	0.7017	-2.203	0.0276 *

```
## Clinical trial data
```

```
trials <- data.frame(  
  Center = factor(rep(1:5, each = 2)),  
  Treatment = factor(rep(c("Active drug", "Placebo"), 5)),  
  Success = c(0, 0, 1, 0, 0, 0, 6, 2, 5, 2),  
  Failure = c(5, 9, 12, 10, 7, 5, 3, 6, 9, 12)  
)  
trials
```

```
trials_model_logistic <- glm(cbind(Success, Failure) ~  
  Center + Treatment -1, data = trials, family =  
  binomial(link=logit))  
summary(trials_model_logistic)
```

Effects of Sparse Data (Cont'd)

(Practice) Clinical Trial Data

Interpretation

There are 0 counts in center 1 and 3, which makes ASE of MLEs large

If parameter estimation procedure does not converge, we can remove center 1 and 3 from the dataset or pool those with other centers

Firth's biased-reduced penalized-likelihood logistic regression

$$l^*(\beta) = l(\beta) + \frac{1}{2} \log |I(\beta)|$$

Model fitted by Penalized ML

Coefficients:

	coef	se(coef)	lower 0.95	upper 0.95	Chisq	p	method
Center1	-2.6813197	1.4349899	-7.5477195	-0.5767967	7.1801287	0.0073715418	2
Center2	-2.2772773	0.8476984	-4.4870869	-0.8817633	12.4164149	0.0004255764	2
Center3	-2.8287196	1.4361010	-7.6892955	-0.7578365	8.7596323	0.0030797114	2
Center4	0.5299017	0.5682095	-0.5797049	1.7334405	0.8683557	0.3514109246	2
Center5	-0.4581499	0.4832136	-1.4635140	0.4858718	0.8984407	0.3432001645	2
TreatmentPlacebo	-1.3924143	0.6282041	-2.7929454	-0.1606091	4.9543487	0.0260251257	2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

```
require(logistf)
trials_long <- trials |>
  tidyr::uncount(Success + Failure, .id = "obs") |>
  dplyr::mutate(Outcome = dplyr::if_else(obs <= Success, 1, 0)) |>
  dplyr::select(Center, Treatment, Outcome)

trials_model_logistic_firth <- logistf(Outcome ~ Center +
  Treatment-1, data = trials_long)
summary(trials_model_logistic_firth)
```

Adjusting sampling bias in oversampled logistic modeling

Case-control studies

If the probability of observing event is very small, case-control studies are efficient and improve the statistical power. However, there is a sampling bias, and there are some biases for regression coefficients

Solution

$$\text{Model } \log\left(\frac{P(y_i = 1 | x_i)}{1 - P(y_i = 1 | x_i)}\right) = \beta_0 + \beta_1 x_i$$

β_1 : unbiased

β_0 : biased, and if we denote the proportion of case in the population and samples by τ and α , the unbiased estimates becomes

$$\hat{\beta}_0 - \log\left[\left(\frac{1 - \tau}{\tau}\right)\left(\frac{\alpha}{1 - \alpha}\right)\right]$$

Adjusting sampling bias in oversampled logistic modeling (Cont'd)

Proof

S: occurrence of an event, F: occurrence of a non-event, A: being ascertained as a sample

α : probability of an event in sample, τ : probability of an event in population

$$\begin{aligned}\log \left[\frac{P(S|A)}{1 - P(S|A)} \right] - \log \left[\frac{P(S)}{1 - P(S)} \right] &= \log \left(\frac{\alpha}{1 - \alpha} \right) - \log \left(\frac{\tau}{1 - \tau} \right) \\ &= \log \left(\frac{(1 - \tau)}{\tau} \frac{\alpha}{(1 - \alpha)} \right) \\ \Leftrightarrow \beta_0^P + \beta_1^P x_i &= \beta_0^S + \beta_1^S x_i - \log \left[\frac{1 - \tau}{\tau} \frac{\alpha}{1 - \alpha} \right]\end{aligned}$$

Exact test for logistic regression

Sufficient statistics

$$\text{Model} \quad \log\left(\frac{P(y_i = 1 | x_i)}{1 - P(y_i = 1 | x_i)}\right) = \beta_0 + \beta_1 x_i$$

$$\text{Sufficient statistics for } (\beta_0, \beta_1): \quad T \equiv (T_0, T_1) \equiv \left(\sum_{i=1}^n y_i, \sum_{i=1}^n x_i y_i \right)$$

$$\begin{aligned} P(T_0 = t_0, T_1 = t_1) &= \sum_{\left\{ y_i^* : \sum_{i=1}^n y_i^* = t_0, \sum_{i=1}^n x_i y_i^* = t_1 \right\}} \frac{\exp(\beta_0 y_i^* + \beta_1 x_i y_i^*)}{\prod_{i=1}^n (1 + \exp(\beta_0 + \beta_1 x_i))} \\ &= \frac{C(t_0, t_1) \exp(\beta_0 t_0 + \beta_1 t_1)}{\prod_{i=1}^n (1 + \exp(\beta_0 + \beta_1 x_i))} \end{aligned}$$

Exact test for logistic regression (Cont'd)

Conditional logistic regression

Conditional probability

$$P(T_1 = t_1 | T_0 = t_0) = \frac{P(T_1 = t_1, T_0 = t_0)}{P(T_0 = t_0)} = \frac{C(t_0, t_1) \exp(\beta_1 t_1)}{\sum_{t_1^*} C(t_0, t_1^*) \exp(\beta_1 t_1^*)}$$

Example

Observation	y	x_0	x_1
1	0	1	1
2	1	1	1
3	0	1	2
4	1	1	0

Exact test for logistic regression (Cont'd)

Example

$$t_0 = \sum_{i=1}^4 y_i = 2, t_1 = \sum_{i=1}^4 x_i y_i = 1$$

Vector	y_1	y_2	y_3	y_4	t_0	t_1
1	0	0	0	0	0	0
2	0	0	0	1	1	0
3	0	0	1	0	1	2
4	0	0	1	1	2	2
5	0	1	0	0	1	1
6	0	1	0	1	2	1
7	0	1	1	0	2	3
8	0	1	1	1	3	3
9	1	0	0	0	1	1
10	1	0	0	1	2	1
11	1	0	1	0	2	3
12	1	0	1	1	3	3
13	1	1	0	0	2	2
14	1	1	0	1	3	2
15	1	1	1	0	3	4
16	1	1	1	1	4	4

Exact test for logistic regression (Cont'd)

Example

Conditional distribution can be derived from this joint distribution by extracting every vector with $t_0 = 2$

t_0	t_1	Frequency	Probability under $\beta_1 = 0$
2	1	2	2/6
2	2	2	2/6
2	3	2	2/6
Total		6	1

If there are 30 different observations, we need to consider 2^{30} cases. This is computationally very intensive.

Exact test for logistic regression (Cont'd)

(Practice) A small dose-response study data

```
> dose
```

	Dose	Deaths	Total
1	0	0	3
2	1	0	3
3	2	0	3
4	3	0	3
5	4	1	3
6	5	2	3

Results:

	estimate	p-value	p-value_se	mc_size
Dose	1.86493	0.0118	0.00164	15000

95% Confidence Intervals for Parameters

	lower	upper
Dose	0.1420113	5.992394

```
## Exact test for logistic regression
```

```
require(elrm)
```

```
dose <- data.frame(
```

```
  Dose   = c(0, 1, 2, 3, 4, 5),
```

```
  Deaths = c(0, 0, 0, 0, 1, 2),
```

```
  Total  = c(3, 3, 3, 3, 3, 3)
```

```
)
```

```
dose
```

```
dose_model_elrm <- elrm(
```

```
  formula = Deaths/Total ~ Dose,
```

```
  interest = ~ Dose, r = 2, iter = 20000, burnIn =
```

```
5000, dataset = dose
```

```
)
```

```
summary(dose_model_elrm)
```

Exact test for logistic regression (Cont'd)

The unconditional asymptotic and conditional exact results produce somewhat conflicting conclusions for this example. Stokes, Davis, and Koch (1995) recommended looking at the exact results when sample sizes are small and the approximate p-values are less than 0.10. For this example, the small sample size and the conflicting results for the asymptotic hypothesis tests indicate that an exact analysis would be more appropriate. Stokes, Davis, and Koch (1995) observe that, in general, the exact methods tend to produce more conservative results.